

Project Title

E-Commerce Product Entity Resolution & Matching Platform

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **E-Commerce Product Entity Resolution & Matching Platform** is a Data Structures and Algorithms (DSA)-based project that identifies and matches identical products listed by different online sellers. In online marketplaces, the same product is often presented with different names, descriptions, and prices, making it difficult for customers to compare products accurately. This platform addresses the problem by collecting product information from multiple sellers, normalizing the product data, and comparing key attributes such as brand, model, storage, color, and specifications to determine whether different listings refer to the same product.

The system uses efficient DSA concepts such as **HashMap**, **HashSet**, **ArrayList**, searching, sorting, and string matching algorithms to group duplicate products into a single canonical product entry. It also provides a price comparison feature that displays all available sellers and highlights the lowest price for the matched product. Additionally, users can search products, filter results, and sort listings based on price, brand, or seller.

This project demonstrates the practical application of DSA in solving real-world e-commerce problems without requiring machine learning or complex databases. It is implemented using **Java** and is suitable for academic mini-projects. The platform improves product organization, reduces duplicate listings, enhances the shopping experience, and helps users make informed purchasing decisions through accurate product matching and comparison.

Signature of the Guide